

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (ME) Examination

Nov-Dec 2018

ME403.01 Production & Operation Management**Date: 26.11.2018, Monday****Time: 01.30 p.m. to 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q-1 State True or False. (Each carries one mark)****[07]**

- (1) Production management involves application of planning, organizing, directing and controlling the production process.
- (2) Production Management is a set of general principles for facility design, job design, schedule design, quality control, inventory control, work study and budgetary control.
- (3) The right quality is not necessarily best quality. It is determined by the cost of the product and the technical characteristics as suited to the specific requirements.
- (4) Power plant's capacity is measured in megawatts of electricity produced.
- (5) CPM uses three time estimates to calculate activity time.
- (6) CPM is useful where activities are subjected to considerable degree of uncertainty in performance time.
- (7) Indirect materials are the raw materials that become part of the product.

Q-2 Answer following questions. (Any two)**[14]**

- (1) What is meant by product design? What are durability and reliability with respect to product design?
- (2) What are the objectives of production planning and control? What is the connection between production planning and production control?
- (3) What do you understand from capacity planning? Which activities involved in capacity planning?

Q-3 Answer following questions. (Any two)**[14]**

- (1) There are seven jobs, each of which has to go through the machines A and B in the order AB. Processing time in hours are given as-

Job	1	2	3	4	5	6	7
Machine A	3	12	15	6	10	11	9
Machine B	8	10	10	6	12	1	3

Determine a sequence of these jobs that will minimize the total elapsed time. Also find idle time for machine A and B.

- (2) Explain the rules of network construction and differentiate between CPM and PERT.
- (3) What is Bill Of Material (BOM)? Which steps are involved in Material Requirement Planning (MRP)?

SECTION – II

Q-4 State True or False. (Each carries one mark)

[07]

- (1) Short range forecasts are less accurate than long range forecasts.
- (2) Johnson's rule is one of the forecasting techniques.
- (3) Delphi method is used for Judgmental forecast.
- (4) The ISO 14000 standards provide a guideline or framework for organizations that need to systematize and improve their environmental management efforts.
- (5) Benchmarking determines how company is doing relative to others.
- (6) Precision in production does not mean parts produced have little or no variation.
- (7) Low cost, higher volume items requires 100% inspection.

Q-5 Answer following questions. (Any two)

[14]

- (1) Give classification of production systems and write down the factors which influence the choice of manufacturing system.
- (2) Suppose a firm is manufacturing automobiles and finds a relationship between sale of automobile and the index demand for cars. The sales for the last five years are-

Year	2013	2014	2015	2016	2017
Sales	110	130	150	160	180
Demand Index	100	110	140	150	200

Find the relation between the demand index and sale of automobile. Further, make a forecast for the year 2018, supposing the demand index rise to 210. (Use least square method)

- (3) The annual sales of an enterprise are as follows-

Years	2013	2014	2015	2016	2017
Sales in lacs of Rs.	45	56	78	46	75

By adopting the method of least squares, find the trend values for each of the five years. Also estimate the annual sales for 2018.

Q-6 Answer following questions. (Any two)

[14]

- (1) Define O.C. curves and explain its significance.
- (2) Explain in brief-
 - (a) ISO 9000
 - (b) Total Quality Management
- (3) What is Poka-Yoke? Write down the different devices or methods used in it.
